

Damien Robert

Deep Learning for Remote Sensing & Environment



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I am a postdoctoral researcher in the [EcoVision lab](#) at [University of Zurich](#), I collaborate with [Jan Dirk Wegner](#) to design deep learning methods for remote sensing and environmental applications. Before joining UZH, I completed my PhD on "[Efficient Learning on Large-Scale 3D Point Clouds](#)" at [IGN](#) and [ENGIE](#), under the supervision of [Loïc Landrieu](#) and [Bruno Vallet](#).

■ POSITIONS

- | | |
|-----------|---|
| 2024–now | Postdoctoral Researcher
EcoVision lab , University of Zurich · Zurich, Switzerland
Deep learning for remote sensing and environment
PI: Jan D. Wegner |
| 2020–2024 | PhD Student
ENGIE Lab CRIGEN — LASTIG , IGN/ENSG · Paris, France
Efficient learning on large-scale 3D point clouds
Advisors: Loïc Landrieu , Bruno Vallet |
| 2017–2020 | R&D Engineer
SIRADEL , ENGIE · Rennes, France
Deep learning on large-scale terrestrial/aerial, indoor/outdoor 3D and 2D data |
| 2017 | Co-Founder
Inspirama
Website gathering book recommendations from inspiring people |
| 2015 | R&D Intern
Dassault Systèmes
Dimensionality reduction and dynamic system modeling |

■ EDUCATION

- | | |
|-----------|---|
| 2011–2015 | MSc, Engineering
École Centrale de Lyon · Lyon, France
Mathematics, Computer Science, Mechanics, Signal Processing, Automation |
| 2022 | International Computer Vision Summer School
ICVSS · Sicily, Italy
Computer vision courses by world-renowned experts in academia and industry |
| 2017 | CNRS AI Fall School
CNRS · Lyon, France
Multi-disciplinary course for AI students and researchers |

■ AWARDS

- | | |
|------|--|
| 2026 | Outstanding Reviewer · CVPR |
| 2025 | PhD Award — accessit · AFRIF
French Association for Pattern Recognition and Interpretation |
| 2024 | Outstanding Reviewer · ECCV |
| 2022 | Best Paper Finalist · CVPR
Top 0.4% of submissions |

2026

Bias Aware Benchmark for Species Distribution Modeling

E. Arens, N. van Tiel, R. Zbinden, C. Vanalli, **D. Robert**, L. Drees, B. Kellenberger, L. Pellissier, J. D. Wegner, D. Tuia
[World Biodiversity Forum](#)

Opportunities and challenges for near-term ecological forecasting with biodiversity and EO data

P. Singh, B. Bektas, S. Evers, E. Fenollosa, G. Koren, S. K. M. Parvathi, S. Pang, M. Paniw, **D. Robert**, G. Sialelli, J. Slingsby, R. Thornley, E. L. Underwood, J. D. Wegner, W. Wu
[World Biodiversity Forum](#)

LitePT: Lighter Yet Stronger Point Transformer

Y. Yue, **D. Robert**, J. Wang, S. Hong, J. D. Wegner, C. Rupprecht, K. Schindler
[CVPR](#) · [Paper](#) · [Project](#) · [Code](#)

Highlight

EZ-SP: Fast and Lightweight Superpoint-Based 3D Segmentation

L. Geist, L. Landrieu, **D. Robert**
[ICRA](#) · [Paper](#) · [Project](#) · [Code](#)

FORMSpot: Revealing fine-scale forest disturbances from nation-wide 1.5 m forest canopy height time series

M. Schwartz, F. Fogel, N. Besic, **D. Robert**, L. Geist, J. Renaud, J. Monnet, C. Mosig, C. Vega, A. d'Aspremont, L. Landrieu, P. Ciais
[Remote Sensing of Environment](#) · [Paper](#)

2025

MT-GSR4B: Multi-Temporally Guided Super-Resolution for Above-Ground Biomass Estimation

K. Karaman, V. S. F. Garnot, **D. Robert**, M. J. Santos, J. D. Wegner
[PFG – Journal of Photogrammetry, Remote Sensing and Geoinformation Science](#) · [Paper](#) · [Code](#)

SSL4Eco: A Global Seasonal Dataset for Geospatial Foundation Models in Ecology

E. Plekhanova, **D. Robert**, J. Dollinger, E. Arens, P. Brun, J. D. Wegner, N. Zimmermann
[CVPR workshop EarthVision](#) · [Paper](#) · [Project](#) · [Code](#)

GSR4B: Biomass Map Super-Resolution with Sentinel-1/2 Guidance

K. Karaman, Y. Jiang, **D. Robert**, V. S. F. Garnot, M. J. Santos, J. D. Wegner
[ISPRS Geospatial Week](#) · [Paper](#) · [Project](#) · [Code](#)

Oral

Climplicit: Climatic Implicit Embeddings for Global Ecological Tasks

J. Dollinger, **D. Robert**, E. Plekhanova, L. Drees, J. D. Wegner
[ICLR workshop Tackling Climate Change with Machine Learning](#) · [Paper](#) · [Project](#) · [Code](#)

Lossy Neural Compression for Geospatial Analytics: A Review

C. Gomes, I. Wittmann, **D. Robert**, J. Jakubik, T. Reichelt, M. Martone, S. Maurogiovanni, R. Vinge, J. Hurst, E. Scheurer, R. Sedona, T. Brunschweiler, S. Kesselheim, M. Batic, P. Stier, J. D. Wegner, G. Cavallaro, E. Pebesma, M. Marszalek, M. A. B. Plomer, K. Adriko, P. Fraccaro, R. Kienzler, R. Briq, S. Benassou, M. Lazzarini, C. M. Albrecht
[IEEE Geoscience and Remote Sensing Magazine](#) · [Paper](#)

2024

Efficient Learning on Large-Scale 3D Point Clouds

D. Robert
[PhD Thesis](#), Université Gustave Eiffel

Jury: Sébastien Lefèvre, Cédric Demonceaux, Patrick Pérez, Siyu Tang, Duygu Ceylan, Loic Landrieu, Bruno Vallet ·
[Paper](#) · [Code](#)

AFRIF award

Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering

D. Robert, H. Raguet, L. Landrieu
[3DV](#) · top 5.3% of submissions · [Paper](#) · [Project](#) · [Code](#)

Oral

2023

Efficient 3D Semantic Segmentation with Superpoint Transformer

D. Robert, H. Raguet, L. Landrieu
[ICCV](#) · top 26.8% of submissions · [Paper](#) · [Project](#) · [Code](#)

2022

Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation

D. Robert, B. Vallet, L. Landrieu
[CVPR](#) · top 0.4% of submissions · [Paper](#) · [Project](#) · [Code](#)

Oral · Best paper finalist

UNDER REVIEW

2026	Toward a Foundation Model for Forest Point Clouds Y. Yue, S. Puliti, D. Robert , A. Topaloglu, B. Xiang, M. Wielgosz, J. D. Wegner, R. Astrup, C. Rupprecht, K. Schindler
2026	From Machine Learning to Large-Scale EO Products: Best Practices for Making Maps G. Sialelli, R. Young, Y. Jiang, C. Aybar, L. Scheibenreif, D. Robert , C. Mosig, A. J. Stewart, J. D. Wegner, A. Pirinen, O. Mogren, K. Schindler

■ COLLABORATION & SUPERVISION

PhD Students	Emilia Arens UZH with Jan D. Wegner · Johannes Dollinger UZH with Jan D. Wegner · Kaan Karaman UZH with Jan D. Wegner · Louis Geist ENPC with Loïc Landrieu · Yuanwen Yue ETH / Oxford with Konrad Schindler and Christian Rupprecht
Master Students	Valerio Schelbert ETH · Jackson Sunny École Polytechnique with SAMP
Collaborators	Konrad Schindler ETH · Christian Rupprecht Oxford · Jan D. Wegner UZH · Loïc Landrieu ENPC · Vivien Sainte Fare Garnot UZH · Lukas Drees UZH · Martin Schwartz LSCE · Elena Plekhanova WSL
Advising	UpCircle AI Switzerland · École des Sciences Criminelles de Lausanne Switzerland · SAMP AI France · GEODIGITAL US

■ REVIEWING & CHAIRING

2026	ECCV Area Chair · CVPR — Outstanding Reviewer
2025	CVPR · CVPR workshop EarthVision
2024	ECCV — Outstanding Reviewer · ISPRS Journal of Photogrammetry and Remote Sensing · ICLR workshop ML4RS · CVPR workshop EarthVision
2023	CVPR · CVPR workshop EarthVision
2022	ISPRS Journal of Photogrammetry and Remote Sensing · CVPR workshop EarthVision

■ TEACHING

2026	Transformer · UZH · Lecture, Master 2	
2025	AI4Good group projects · UZH · Labs, Master 2	14 h
2025	Transformer · UZH · Lecture, Master 2	4 h
2024	Transformer, NeRFs, and Diffusion · UZH · Lecture and labs, Master 2	5.5 h
2023	Deep Learning for Remote Sensing · ENSG-IGN · Lecture, Master 2	13 h
2022	3D Deep Learning for Remote Sensing · ISPRS'22 · Tutorial, Researchers	1 day
2022	3D Deep Learning, Torch-Points3D & DeepViewAgg · ENGIE CRIGEN lab · Tutorial, Researchers	1 day
2022	Deep Learning for Remote Sensing · ENSG-IGN · Lecture, Master 2	9 h
2020	Deep Learning for Computer Vision · Ecole Polytechnique · Lecture, Master 1	12 h

■ OPEN-SOURCE SOFTWARE

★ 405	LitePT · prs-eth/LitePT	44 forks
★ 15	torch-graph-components · drprojects/torch-graph-components	
★ 1054	Superpoint Transformer · drprojects/superpoint_transformer	136 forks
★ 53	NoRA · drprojects/nora	3 forks
★ 74	Point Geometric Features · drprojects/point_geometric_features	8 forks
★ 237	DeepViewAgg · drprojects/DeepViewAgg	24 forks

■ SELECTED TALKS

2026	ESA-NASA workshop on AI Foundation Model for EO · tutorial · Building Earth Observation Embedding Workflows with TerraTorch · Virtual
	ESA Φ-lab · Geospatial foundation models for forest disturbance monitoring · Frascati, Italy
2025	MPI of Animal Behavior · Research at EcoVision lab · Zurich, Switzerland
	Global Forest Watch · Geospatial foundation models for forest disturbance monitoring · Virtual
	NatureFinance · Research at EcoVision lab · Virtual
	NVIDIA Strategic Research Engagement · Superpoint Transformer and EcoVision research · Zurich, Switzerland
	CNES COMET TSI · Remote Sensing for Species Distribution Modeling · Virtual
2024	3D Data Academy · interview · Interview by Florent Poux and tutorial for Superpoint Transformer · YouTube
	Google DeepMind · Research at EcoVision lab · Zurich, Switzerland
	3DV'24 · oral · Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering · Davos, Switzerland
2023	Ecole des Ponts, IMAGINE lab · Efficient Learning on Large-Scale 3D Point Clouds · Paris, France
	National Land Survey of Finland (NLS) · IGN's research on Large-Scale 2D and 3D Learning · Paris, France
	ETH Zürich, Computer Vision and Geometry lab · Efficient Learning on Large-Scale 3D Point Clouds · Zurich, Switzerland
	ETH Zürich, Photogrammetry and Remote Sensing lab · Efficient Learning on Large-Scale 3D Point Clouds · Zurich, Switzerland
	Valeo.ai · Efficient 3D Semantic Segmentation with Superpoint Transformer · Paris, France
2022	CV News · interview · Interviewed by the CV News journal for its Best of CVPR'22 issue · New Orleans, US
	CVPR'22 · oral · Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation · New Orleans, US
	45 talks given since 2021 — full list at drprojects.github.io/talks .

■ SKILLS & LANGUAGES

Research	3D & geometry Large-scale 3D point clouds · LiDAR data · Superpoint-based learning · Graph-based learning and graph partitioning
	Efficient & scalable learning Efficient deep learning · Scalable model design · EO data representation and compression
	Learning paradigms Multimodal learning · Self-supervised learning · Representation learning
	Environment & Earth observation Remote sensing · Forest structure analysis · Species distribution modeling · Representation learning for macroecology
Tools	Python · PyTorch · PyTorch Lightning · PyTorch Geometric · C++ · Hydra · Weights & Biases · Git · Linux · SLURM · scikit-learn · Plotly · Blender · LaTeX
Languages	French Native
	English Fluent
	Spanish Intermediate
	German Beginner
Abroad	Zurich, Switzerland 2024–now · Backpacking, Worldwide 2015–2016 · Providence, RI, United States 2014–2015
Interests	Nature & forests · Boxing · Mountaineering · Skateboarding · Guitar · Painting · Reading · Board games